

Analog Electronic Circuits

A Simplified Approach

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Sanguine Technical Publishers

Bangalore - 560016

2014

Price: Rs. 375.00
USD \$ 24.99



Title: Analog Electronic Circuit: A Simplified Approach
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© 2008 by Sanguine Technical Publishers, Bangalore – 560 016.

Published by *Lal M Prasad* for *SANGUINE*

Production Editor: *R.Subramanian*

Typeset in *Times New Roman* by *Sigma Publishing Solutions, Chennai – 600 033*.

*Submitted at the Lotus Feet of
His Holiness
Parama Poojya Lingaikya Jagadguru
Shree Shree Shree Shivaratri Rajendra Mahaswamigalavaru,
Founder President
J. S. S. Mahavidyapeeta
Mysore.*

*In Fond Memory of My Grand Mother
Late Smt. M. S. Basamma
and
My Father
Late Shree T. N. Basavaraju
who are responsible for what I am today.*

FOREWORD

I am extremely happy to note that Mr. U. B. Mahadevaswamy from Sri Jayachamarajendra College of Engineering, Mysore has authored a book on ANALOG ELECTRONIC CIRCUITS which is helpful to the students of all Electrical Science branches of Engineering.

He is one among the most admired teacher of the Electronics & Communication department of SJCE, Mysore.

With his immense popularity as an excellent teacher among the student community for his dedicated teaching, I am sure the book will be of great asset to the students.

With his rich teaching experience of nearly two decades, this book being brought out, is a boon to the student community.

I would definitely recommend this book to not only the student community but to anyone who wants to advance their knowledge in Analog Electronics.

Dr. M. Shanthakumar

A handwritten signature in dark ink, appearing to read 'M. Shanthakumar', with a horizontal line underneath the name.

Date: August 13, 2008

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MAURITIUS

PREFACE

This book is the fruit of my rich experience of teaching the subject 'Analog Electronics' nearly for a span of 18 years. The various concepts of the subject are arranged logically and explained in a simple reader friendly language for proper understanding of the subject. A large number of problems with their step by step solution are provided for every concept. Illustrative examples are discussed to emphasize on conceptual clarity thereby presenting typical applications.

This book takes you from simple diode circuits through the analysis and design of a variety of transistor amplifiers. Analysis and design of practical feedback amplifiers and various oscillator circuits have also been covered. This book concludes with FET amplifiers wherein various FET configurations, their analysis and design have been discussed.

This book provides a simplified and systematic approach to difficult theoretical concepts in Analog Electronics. It can serve as an excellent reference material for design engineers. Those of you who enjoyed reading our previous book entitled 'Electronic Circuits' would certainly enjoy this book too.

Despite the delight taken by many reviewers in finding mistakes a few typo errors have managed to slip through the sieve. This suggests that more await discovery by you. I suppose that is what second editions are for.

U. B. MAHADEVASWAMY

ACKNOWLEDGMENTS

My sincere Pranamas at the Lotus Feet of His Holiness Parama Poojya Jagadguru Shree Shree Shree Shivaratri Deshikendra Mahaswamigalavaru, President, J.S.S Mahavidyapeeta, Mysore, whose divine blessings inspired me to take up and successfully complete this project.

A special word of thanks to Prof. M.H.Dhananjaya, Director (Technical), who continues to be an infinite source of encouragement in all my endeavours.

I gratefully acknowledge all the encouragement from Dr. B.G.Sangameshwara, Principal, SJCE, Mysore in providing all the facilities and resources in drafting this book.

My special thanks to my teacher and my well wisher Dr.M.Shantakumar, Principal, JSS Academy of Technical Education, Mauritius, who motivated me to take up this venture and encouraged me throughout this project.

I must acknowledge the support and encouragement rendered by Prof. Dr. B. S. Basavarajiah, Special Officer, SJCE, Mysore during this project.

I sincerely thank my beloved H.O.D, Dr. R. D. Sudhaker Samuel, who encouraged me to take up this project. His support and invaluable suggestions at each and every stage of drafting this book are unforgettable. I hope without his support and encouragement, this book would hardly be in its present form.

My sincere thanks to my colleague Mr. V. Nattarasu and Mr. R. Subramanian of Sanguine Technical Publishers, who have meticulously reviewed this book and gave invaluable inputs to bring this book to its present form.

I gratefully acknowledge the support rendered during this project, by my teacher and well wisher Mr. C.Chamaraju, Assistant professor, Mathematics department, SJCE, Mysore.

Thanks to Dr.Ganesh Rao, Professor, Department of Telecommunication, MSRIT, Bangalore, for coming to my rescue at various stages in the preparation of this book.

I must thank my colleagues in the department of Electronics and Communication, SJCE, Mysore for all their suggestion towards improving the material presented.

I must thank Ms. Sheethal M. J and Ms. Swetha M. J, software Engineers, for their timely technical inputs, support and encouragement throughout this project.

I appreciate the relentless efforts of my sister's children Mr. G. Shashidharamurthy, software Engineer and Ms. Indumathi, in finding mistakes and typo errors through several readings. I also appreciate the help rendered by Ms. K. S. Sindhu, who patiently checked the solutions of all numerical examples.

I must acknowledge all the encouragement and support from my wife K. S. Umadevi in making this project a success. The timely support of my brother B.Chandra Shekara Murthy, Quality Control Engineer, Muscat and my mother K.P.Leelavthi and all my family members needs a very special mention.

Finally, my sincere gratitude for Sanguine Publishers for continuing to make my dreams a reality.

Please feel free to contact the publisher at info@sanguineindia.com for further assistance. I am sincerely grateful to all the persons mentioned above for their generous support at the right time to make this work truly marvelous.

U. B. MAHADEVASWAMY

CONTENTS

Diode Circuits

1.1	Volt-ampere Characteristics of Semiconductor Diode	1
1.2	Temperature Effects	4
1.3	Comparative Study of Ge, Si and GaAs Devices	6
1.4	Diode Resistance Levels	8
1.5	Analytical Expression for Dynamic Resistance of Diode	13
1.6	Equivalent Circuits of Diode	16
1.7	Transition and Diffusion Capacitance	18
1.8	Charge Storage and Reverse Recovery Time	20
1.9	Diode Specification Sheets	21
1.10	Classification of Junction Diodes	22
1.11	Load-line Analysis of Diode Circuit	22
1.12	Analysis of Diode Circuits using the Diode Equivalent Circuit	24
1.13	Rectifiers	45
1.14	Half-wave Rectifier	45
1.15	Full-wave Bridge Rectifier	57
1.16	Practical Rectifier Circuits	67
1.17	Full Wave Rectifier using Two Diodes and a Center-trapped Transformer	68
1.18	Analysis of Rectifier Circuits considering the Effect of Diode Resistance	72
1.19	Clipping Circuits or Limiting Circuits	73
1.20	Shunt or Parallel Clipper	73
1.21	Series Clipper	76
1.21	Clipping at Two Independent Levels [Double Ended Clipper]	79
1.22	Another Double Ended Clipper	82
1.23	Clamping Circuits	113
1.24	Negative Clamper	113
1.25	Positive Clamper	116

Transistor Biasing

2.1	The Operating Point	129
2.2	DC Load Line	131
2.3	Reasons for Instability of the Operating Point	132
2.4	Stability Factors	133
2.5	Fixed-bias Circuit	134
2.6	Load Line Analysis of the Fixed-bias Circuit	141

2.7	Emitter-bias Circuit	142
2.8	Voltage Divider Bias or Universal Bias Circuit	151
2.9	Collector Feedback Bias	164
2.10	Design Rule	188
2.11	Circuits with <i>pnp</i> Transistors	191
2.12	Transistor Switching Networks	195
2.13	Transistor Switching Times	204
2.14	Bias Stabilisation	205
2.15	General Expression for Stability Factor $S(I_{CO})$	205
2.16	Stability Factors for Fixed-bias Circuit	206
2.17	Stability Factors for Emitter-bias Circuit	209
2.18	Stability Factors for Voltage Divider Bias Circuit	213
2.19	Stability Factors for Collector Feedback Bias Circuit	217

Transistor at Low Frequencies

3.1	Notations in Amplifier Analysis	231
3.2	AC Resistance of Emitter Base Diode	232
3.3	Two Port Network	232
3.4	Modeling of Transistor	233
3.5	The r_e Model of Transistor in Common Base (CB) Configuration	233
3.6	r_e Model of <i>pnp</i> Transistor in CB Configuration	235
3.7	The r_e Model of Transistor in Common-emitter (CE) Configuration	237
3.8	r_e Model of Transistor in CE Configuration including r_o	239
3.9	The Hybrid Equivalent Model	240
3.10	Hybrid Model of CB Configuration	243
3.11	Hybrid Model of CE Configuration	244
3.12	Hybrid Model of CC Configuration	245
3.13	Variations of h -parameters	246
3.14	Advantages of h -parameters	248
3.15	Approximate Hybrid Equivalent Circuit	249
3.16	Relation between the Parameters of Hybrid Model and the r_e Model	250
3.17	Hybrid π Model	253
3.18	Common-emitter Fixed-bias Configuration	257
3.19	Common-emitter Configuration with Voltage Divider Bias	264
3.20	Common-emitter Configuration using Emitter-bias with Unbypassed R_E	269
3.21	CE Emitter Bias Configuration with Bypassed R_E	273
3.22	Common-emitter Configuration using Voltage Divider Bias with Unbypassed R_E	275
3.23	Voltage Divider Bias and Emitter Bias Configurations with Partly Bypassed R_E	277
3.24	Emitter-follower Configuration	278
3.25	Emitter-follower using Voltage Divider Bias	283
3.26	Emitter-follower using Collector Resistor	283
3.27	Common-base Configuration	286

3.28	Important Characteristics of CB Configuration	288
3.29	Applications of CB Configuration	289
3.30	Collector Feedback Configuration	290
3.31	Collector DC Feedback Configuration	295
3.32	Effects of R_S and R_L	299
3.33	Fixed Bias Common-emitter Amplifier with R_S and R_L	300
3.34	Common-emitter Voltage Divider Bias Configuration, with R_S and R_L	305
3.35	Emitter-follower Configuration with R_S and R_L	306
3.36	CE Emitter-bias Configuration with R_S and R_L	308
3.37	Two-port Systems Approach	309
3.38	Analysis of Amplifiers using Two-port System Approach considering the Effects of R_S and R_L	311
3.39	Advantages of Two-port System Approach	312

Transistor Frequency Response

4.1	General Frequency Considerations	317
4.2	Low Frequency Analysis	322
4.3	Low Frequency Response of BJT Amplifier	334
4.4	Miller Effect Capacitance	354
4.5	High-frequency Response of BJT Amplifier	359

General Amplifiers

5.1	Cascaded Systems	377
5.2	Cascode Connection	383
5.3	Darlington Connection	386
5.4	DC Bias of Darlington Circuit	387
5.5	Darlington Emitter-follower	388
5.6	Feedback Pair	394
5.7	DC Bias of Feedback Pair	394
5.8	AC Operation of Feedback Pair	396
5.9	Current Mirror Circuit	401
5.10	Current Source Circuits	406
5.11	BJT Constant-current Source	406
5.12	BJT Constant-current Source using Zener Diode	408
5.13	Analysis of Transistor Configurations using Approximate Hybrid Model	409
5.14	Common-emitter Fixed Bias Configuration	410
5.15	Common-emitter Configuration using Voltage Divider Bias	411
5.16	CE-emitter Bias Configuration with Unbypassed R_E	413
5.17	Emitter-follower Configuration	415
5.18	Common-base Configuration	416
5.19	Analysis of General Transistor Configuration using the Complete Hybrid Equivalent Model	417

Feedback Amplifiers

6.1	Classification of Amplifiers	431
6.2	Feedback Concept	437
6.3	Transfer Gain of Single Loop Feedback Amplifier	442
6.4	Loop Gain	444
6.5	Assumptions in the Analysis of Feedback Amplifiers	445
6.6	Feedback Amplifier Topologies	445
6.7	General Characteristics of Negative Feedback Amplifier	451
6.8	Effect of Negative Feedback on Input Resistance	456
6.9	Effect of Negative Feedback on Output Resistance	461
6.10	Effect of Negative Feedback on Bandwidth	467
6.11	Advantages of Negative Feedback	468
6.12	Practical Feedback Circuits	477

Power Amplifiers

7.1	Classification of Power Amplifiers	503
7.2	Class A Power Amplifier	507
7.3	Harmonic Distortion or Non linear Distortion	520
7.4	Higher-order Harmonic Distortion	527
7.5	Conversion Efficiency	535
7.6	Class B Push-pull Power Amplifier	550
7.7	Class D Power Amplifier	579
7.8	Power Transistor Heat Sinking	580
7.9	Thermal Analogy of a Power Transistor	583

Oscillators

8.1	Basic Principle of Oscillators	587
8.2	Practical Aspects	589
8.3	Classification of Oscillators based on the Elements used in the Feedback Network	589
8.4	RC-phase Shift Oscillator	589
8.5	Transistor RC-phase Shift Oscillator	590
8.6	Merits and Demerits of RC-phase Shift Oscillator	592
8.7	Wein Bridge Oscillator	598
8.8	LC Oscillators (Tuned Oscillators)	604
8.9	Transistor Colpitts Oscillator	605
8.10	Transistor Hartley Oscillator	612
8.11	Crystal Oscillator	620
8.12	Piezo Electric Effect	620
8.13	Characteristics of Quartz Crystals	620
8.14	Crystal Reactance	623

8.15	Transistor Crystal Oscillator	623
8.16	Merits and Demerits of Crystal Oscillator	625
8.17	Applications of Crystal Oscillator	626

FET Amplifiers

9.1	Basics of Junction Field Effect Transistor	629
9.2	FET Current Equation	630
9.3	Transfer Characteristics	630
9.4	Drain Characteristics	631
9.5	FET Small-signal Model	631
9.6	Graphical Determination of g_m	633
9.7	Mathematical Expression for g_m	633
9.8	Relation between I_D and g_m	636
9.9	JFET Common-source Amplifier using Fixed Bias Configuration	636
9.10	JFET Common-source Amplifier using Self-bias Configuration	641
9.11	JFET Common-source Amplifier with Unbypassed R_s	645
9.12	JFET Common-source Amplifier using Voltage Divider Configuration	652
9.13	JFET Common-source Amplifier using Voltage Divider Configuration, with Unbypassed R_s	654
9.14	JFET Source-follower [Common-drain Configuration]	658
9.15	JFET Common-gate Configuration	667
9.16	Depletion-type MOSFETS [D-MOSFET]	676
9.17	Enhancement-type MOSFETS [E-MOSFET]	682
9.18	Expression for Transconductance g_m	682
9.19	Small Signal AC Model of E-MOSFET	682
9.20	E-MOSFET Drain-feedback Configuration	683
9.21	E-MOSFET Common-source Amplifier using Voltage Divider Configuration	690
9.22	Design of FET Amplifiers	692

Bibliography	701
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Index	703
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Chapter 1

DIODE CIRCUITS

Diodes are semiconductor devices which conduct only in one direction. Their behaviour depends on the value and polarity of the applied voltage. This chapter begins with the volt-ampere characteristics of diode and takes the reader through several diode applications. Analysis of each diode circuit is presented, backed by numerous illustrative examples.

◆ 1.1 VOLT-AMPERE CHARACTERISTICS OF SEMICONDUCTOR DIODE

A semiconductor diode results when a junction is formed between a p -type and a n -type semiconductor. Such a p - n junction permits easy flow of charge carriers in one direction but restrains the flow in the opposite direction.

Figure 1.1 shows the biasing arrangement for a p - n junction under both forward and reverse biased conditions.

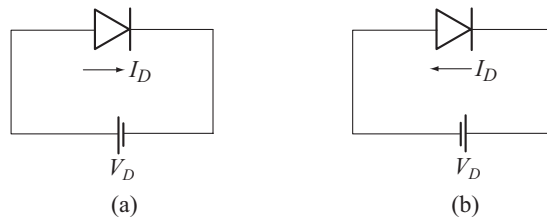


Fig. 1.1 (a) Forward biased p - n junction
(b) Reverse biased p - n junction

The p - n junction is forward biased for $V_D > 0$ and is reverse biased for $V_D < 0$. Under forward bias, p - n junction conducts heavily from anode to cathode while under reverse bias it conducts a small current from cathode to anode.

Diode Current Equation

Through the use of solid-state physics the general characteristics of a semiconductor diode can be defined by the Shockley's equation, for the forward and reverse bias regions given in Equation (1.1).

$$I_D = I_S \left[e^{\frac{V_D}{nV_T}} - 1 \right] \quad (1.1)$$

where I_D = Diode current

V_D = Voltage applied across the diode

I_S = reverse saturation current

n is an ideality factor which depends on the operating conditions and physical construction. It lies between 1 and 2. Unless otherwise stated we assume $n = 1$ throughout the analysis.

V_T = Thermal voltage, given by

$$V_T = \frac{KT}{q}$$

where K = Boltzmann constant = 1.38×10^{-23} J / K

T = Absolute temperature in kelvin

= $273 +$ the temperature in $^{\circ}\text{C}$

q = Magnitude of electronic charge

= 1.6×10^{-19} Coulomb

For example, at a temperature of 27°C

$$T = 273 + 27^{\circ}\text{C} = 300\text{ K}$$

$$\begin{aligned} V_T &= \frac{KT}{q} \\ &= \frac{(1.38 \times 10^{-23})(300)}{1.6 \times 10^{-19}} \\ &\approx 26\text{ mV} \end{aligned}$$

Forward Biased Condition

From Equation (1.1), we have

$$I_D = I_S e^{\frac{V_D}{nV_T}} - I_S \quad (1.2)$$

For positive values of V_D , the exponential term grows very quickly and totally dominates the effect of second term. As a result

$$I_D \approx I_S e^{\frac{V_D}{nV_T}} \quad (1.3)$$

Observe from Equation (1.3) that, under forward biased condition, the diode current varies exponentially with the applied forward voltage V_D .

Reverse Biased Condition

For negative values of V_D , the exponential term in Equation (1.2) drops very quickly below the level of I_S . As a result, the diode current is now given by

$$I_D \approx -I_S \quad (1.4)$$

Note from Equation (1.4) that, for negative values of V_D , the diode current is essentially constant at the level of $-I_S$.

Avalanche Breakdown

Equation (1.4) holds good for small reverse voltages. As the reverse voltage across the diode increases, the velocity of minority carriers responsible for the reverse saturation current I_S will also increase. Eventually, their velocity and the associated kinetic energy will be sufficient to release additional carriers through collisions with otherwise stable atomic structures. An ionisation process will result thereby valence electrons absorb sufficient energy to leave the parent atom. These additional electrons can then aid the ionisation process to the point where a high avalanche current is established and the avalanche breakdown is occurred. The reverse voltage at which the avalanche breakdown occurs is called the reverse breakdown voltage, denoted by V_{BV} or V_{BR} or V_{BD} . In the breakdown region, the diode current increases sharply with diode voltage remaining constant at V_{BD} .

It should be noted that, Avalanche breakdown is a destructive phenomenon. Therefore the diode should not be operated in the avalanche breakdown region. The manufacturer specifies peak reverse voltage (PRV) for the safe operation of diode under reverse bias.

The maximum reverse-bias voltage which can be applied before entering the breakdown region is called peak inverse voltage (PIV rating) or the peak reverse voltage (PRV rating).

For the diode BAY 73, the manufacturer specifies PIV rating of 100 V and break down voltage of 125 V. For this diode if the applied reverse voltage exceeds 100 V, it is likely to enter into break down region.

Volt-ampere Characteristics

Figure 1.2 shows the volt-ampere characteristics of Ge, Si and GaAs diodes. V_K is the knee voltage or the threshold voltage at which the forward current of diode begins to increase exponentially with the applied forward voltage.

Table 1.1 compares the values of V_K , I_S and reverse break down voltages for Ge, Si and GaAs diodes.

Table 1.1 V_K , I_S and V_{BD} for Ge, Si and GaAs diodes

Semi conductor	Parameter		
	V_K	I_S	V_{BD}
Ge	0.3 V	1 μ A	upto 400 V
Si	0.7 V	10 pA	50 V – 20 kV
GaAs	1.2 V	1 pA	50 V – 20 kV

Observe that Ge has large level of I_S and low level of V_{BD} which are highly undesirable. For this reason Ge diodes finds limited application even though V_K is smaller than those of Si and GaAs diodes. Also note that level of V_K for Si is almost one half that of GaAs and the level of V_{BD} is almost same. Hence silicon diodes are more popular.

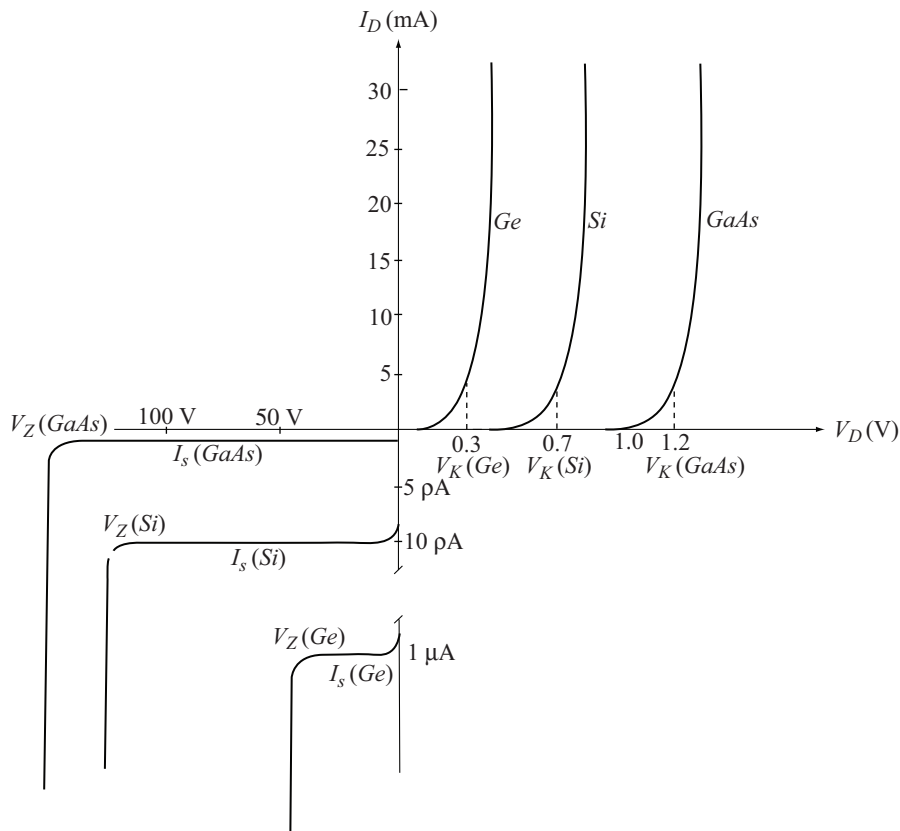


Fig. 1.2 Comparison of Ge, Si, and GaAs diodes

◆ 1.2 TEMPERATURE EFFECTS

The effect of temperature on the characteristics of a silicon diode is illustrated in Fig. 1.3.

In the forward region the characteristics of a silicon diode shift to the left at the rate of 2.5 mV per degree centigrade increase in temperature.

As shown in Fig. 1.3, when the temperature increases from 20°C to 100°C (the boiling point of water), the forward characteristics curve shifts to the left by

$$(100^\circ \text{C} - 80^\circ \text{C}) (2.5 \text{ mV}) = 200 \text{ mV}$$

This shift is significant since the level of V_K for Si is about 0.7 V. A decrease in temperature has the reverse effect as also shown in Fig. 1.3.

In the reverse – bias region the reverse saturation current of a silicon diode doubles for every 10° C rise in temperature.

The doubling effect of I_S with temperature is given by the relation

$$I_S(T_2) = I_S(T_1) 2^{\frac{\Delta T}{10}} \quad (1.5)$$

where

$$\begin{aligned}\Delta T &= \text{Change in temperature} \\ &= T_2 - T_1\end{aligned}$$

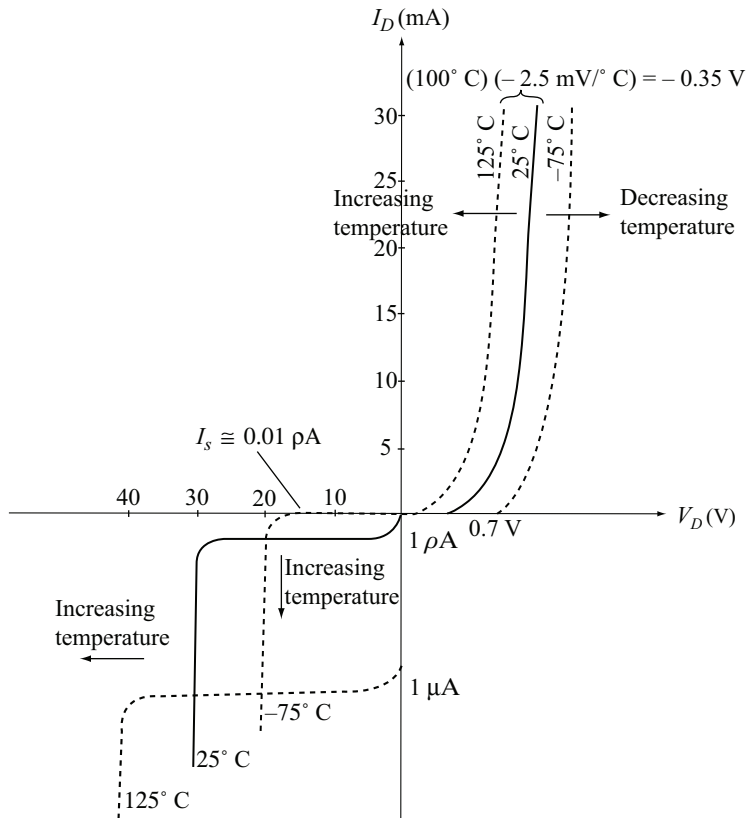


Fig. 1.3 Variation in Si diode characteristics with temperature change

Table 1.2 compares the levels of I_s for Ge, Si and GaAs at an operating temperature of 200°C.

Table 1.2 Level of I_s for Ge, Si and GaAs at 200°C

Operating temperature	Semi conductor		
	Ge	Si	GaAs
20°C	1 μ A	10 pA	1 pA
200°C	262.14 mA	2.62 μ A	0.262 μ A

Observe from table that, an increase in temperature from 20°C to 200°C would result in a monstrous reverse current of 262.14 mA in Ge where as it is only limited to 2.62 μ A in Si and 0.262 μ A in GaAs, due to their excellent temperature characteristics.

Si diodes can work well upto maximum temperature of 200°C and GaAs diodes upto 400°C. Ge diodes are limited to low temperature applications.

The reverse breakdown voltage of a semiconductor diode will increase or decrease with temperature depending on the break down potential.

If the break down voltage is less than 5 V as in the case of zener diodes, the break down voltage decreases with temperature in the same way as shown in Fig. 1.3.

◆ 1.3 COMPARATIVE STUDY OF Ge, Si AND GaAs DEVICES

The comparative study of Ge, Si and GaAs devices with respect to their characteristics and applications is given in Table 1.3.

Table 1.3 Comparative study of Ge, Si and GaAs devices

<i>Semiconductor</i>	<i>Characteristics</i>	<i>Applications</i>
Ge	• High temperature sensitivity • High reverse saturation current • Low break down voltage	• Photo detectors • Security systems
Si	• Good temperature characteristics • Low reverse saturation current • Excellent break down voltage level • Low cost	• Full range of electronic devices • Enormously used in very large scale integrated circuits
GaAs	• Good temperature characteristics • Very low reverse saturation current • Excellent break down voltage levels • High speed of operation • Perhaps the semiconductor material of the future	Major applications in opto electronics • LED • Solar cells • photo detectors • If the manufacturing cost drops, it may be used in integrated circuits, in future.

Example 1.1

Calculate the factor by which the current will increase in a silicon diode operating at a forward voltage of 0.4 V when the temperature is raised from 25°C to 150°C. Take $n = 1$.

Solution

$$T_1 = 25^\circ\text{C} \quad T_2 = 150^\circ\text{C} \quad V_D = 0.4 \text{ V} \quad n = 1$$

$$I_S(T_2) = I_S(T_1) 2^{\frac{\Delta T}{10}}$$

$$\Delta T = T_2 - T_1$$

$$= 150^\circ\text{C} - 25^\circ\text{C} = 125^\circ\text{C}$$

$$\begin{aligned}
 I_S(T_2) &= I_S(T_1) 2^{\frac{125}{10}} \\
 \frac{I_S(T_2)}{I_S(T_1)} &= 5793 \\
 I_D &= I_S \left[e^{\frac{V_D}{V_T}} - 1 \right]
 \end{aligned} \tag{A}$$

Since $n = 1$

$$\begin{aligned}
 I_D &= I_S \left[e^{\frac{V_D}{V_T}} - 1 \right] \\
 I_D(T_1) &= I_S(T_1) \left[e^{\frac{V_D}{V_{T_1}}} - 1 \right]
 \end{aligned} \tag{B}$$

$$I_D(T_2) = I_S(T_2) \left[e^{\frac{V_D}{V_{T_2}}} - 1 \right] \tag{C}$$

Dividing Equation (C) by Equation (B) we have

$$\frac{I_D(T_2)}{I_D(T_1)} = \frac{I_S(T_2)}{I_S(T_1)} \left[\frac{e^{\frac{V_D}{V_{T_2}}} - 1}{e^{\frac{V_D}{V_{T_1}}} - 1} \right] \tag{D}$$

$$V_T = \frac{273 + T}{[q/K]} = \frac{273 + T}{11,600}$$

$$V_{T_1} = \frac{273 + 25}{11600} = 0.0257 \text{ V}$$

$$V_{T_2} = \frac{273 + 150}{11600} = 0.0364 \text{ V}$$

$$e^{\frac{V_D}{V_{T_2}}} - 1 = e^{\frac{0.4}{0.0364}} - 1 = 59218.78$$

$$e^{\frac{V_D}{V_{T_1}}} - 1 = e^{\frac{0.4}{0.0257}} - 1 = 5747078.29$$

Now from Equation (D)

$$\frac{I_D(T_2)}{I_D(T_1)} = 5793 \left[\frac{59218.78}{5747078.29} \right] = 59.69$$

$\Rightarrow$

$$I_D(T_2) = 59.69 I_D(T_1)$$

◆ 1.4 DIODE RESISTANCE LEVELS

The forward characteristics of the diode is non linear since the diode current changes exponentially with the applied forward voltage. As a result the resistance of the diode will change when the operating point moves from one region to another. Depending upon the type of the applied signal we can define the following three resistance levels of the diode.

- (a) DC or static resistance
- (b) AC or dynamic resistance
- (c) Average ac resistance

Now let us study the definition of each of these resistance levels and the procedure to find them from the diode characteristics.

1.4.1 DC or Static Resistance

The application of a dc voltage to a circuit containing a semiconductor results in a dc current. The resistance of diode to the flow of dc current is called DC or static resistance.

Let the applied dc voltage V_D , results in a dc current I_D as shown in Fig. 1.4. The dc resistance of the diode at the operating point is given by

$$R_D = \frac{V_D}{I_D} \quad (1.6)$$

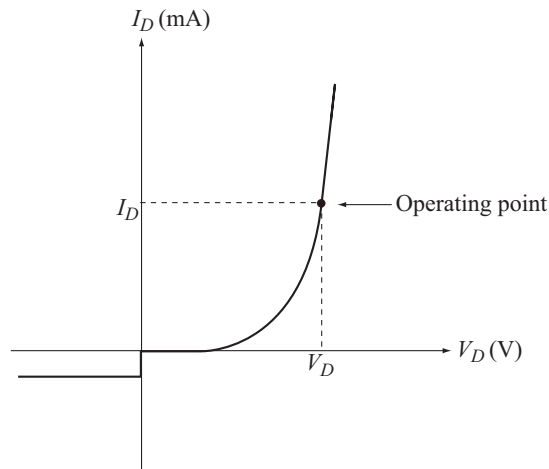


Fig. 1.4 Determination of R_D of a diode

We can make the following observations from Fig. 1.4.

- Under reverse bias, the diode current is almost zero. Hence the dc resistance level in the reverse-bias region is quite high.
- The dc resistance levels at the knee and below will be greater than the resistance levels obtained for the vertical rise section of the characteristics.
- Higher the current through the diode, lower is the dc resistance level.

1.4.2 AC or Dynamic Resistance

The resistance of the diode to the flow of ac current is called the ac or dynamic resistance.

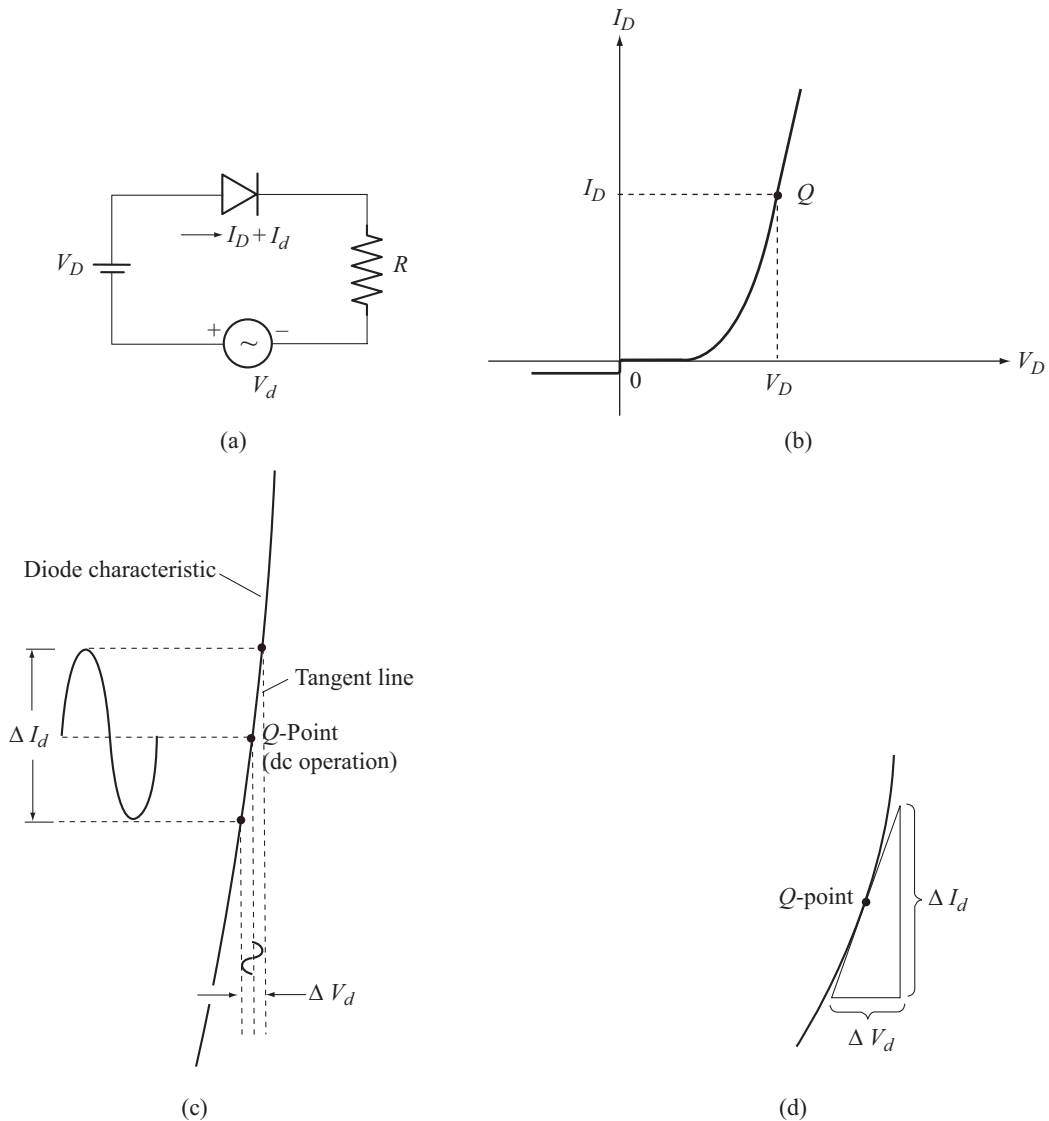


Fig. 1.5 (a) Biased diode with ac input
 (b) Q point current and voltage
 (c) Variation of diode current and voltage about Q point
 (d) Graphical determination of ac resistance

The application of dc voltage V_D in the circuit of Fig. 1.5(a) sets up a dc current I_D resulting in an operating point Q as shown in Fig. 1.5(b). Since V_D and I_D are not changing with time, the operating point is designated as quiescent operating point or Q -point.

If a sinusoidal voltage is also applied then the diode current and voltage will also vary sinusoidally about the Q point as shown in Fig. 1.5(c).

If ΔV_d is the change in ac diode voltage and ΔI_d is the corresponding change in ac diode current, then the ac or dynamic resistance of diode is defined by

$$r_d = \frac{\Delta V_d}{\Delta I_d} \quad (1.7)$$

The level of r_d can be measured by drawing a tangent to the characteristics at the Q point and taking equal increments of diode voltage and diode current about the Q point, as shown in Fig. 1.5(d). The definition of r_d in Equation (1.7) assumes that ΔV_d and ΔI_d are small in amplitude. For this reason r_d is also called the small signal resistance.

It should be noted that, r_d is given by the reciprocal of the slope of the characteristics at the desired Q point. If the Q point is located higher up the curve, the slope becomes steeper [large] resulting in a small value of r_d . Therefore, *the higher the Q point of operation (higher current or higher voltage), the smaller is the ac resistance.*

1.4.3 Average AC Resistance

The ac or dynamic resistance is measured under small signal operation. If the ac input signal amplitude is large it produces a broad swing about the Q point as shown in Fig. 1.6. The resistance associated with the device under large signal operation is called the average ac resistance.

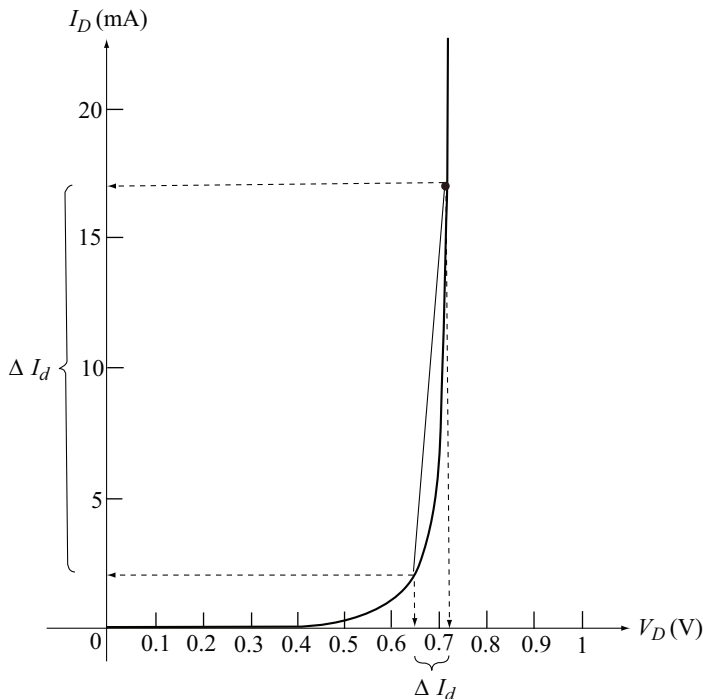


Fig. 1.6 Determining the average ac resistance between indicated limits

The average ac resistance is the resistance determined by a straight line drawn between the two intersections established by the maximum and minimum values of input voltage as shown in Fig. 1.6.

$$r_{av} = \left. \frac{\Delta V_d}{\Delta I_d} \right|_{\text{pt. to pt.}} \quad (1.8)$$

Average ac resistance gives one resistance level for a broad swing of current and voltage levels in the diode. r_{av} is used in the diode equivalent circuits, which are useful in the analysis of diode circuits. *As with the dc and ac resistance levels, the lower the levels of currents used to determine r_{av} , the higher is the resistance levels.*

Example 1.2

Following data are available for a silicon diode.

V_D	-12 V	0.6 V	0.8 V
I_D	1 μA	2.5 mA	30 mA

Calculate the dc resistance levels at

- (a) $I_D = 2.5 \text{ mA}$
- (b) $I_D = 30 \text{ mA}$
- (c) $V_D = -12 \text{ V}$

Solution

dc resistance, $R_D = \frac{V_D}{I_D}$

- (a) At $I_D = 2.5 \text{ mA}$, $V_D = 0.6 \text{ V}$

$$\begin{aligned} R_D &= \frac{0.6 \text{ V}}{2.5 \text{ mA}} \\ &= 240 \Omega \end{aligned}$$

- (b) At $I_D = 30 \text{ mA}$, $V_D = 0.8 \text{ V}$

$$\begin{aligned} R_D &= \frac{0.8 \text{ V}}{30 \text{ mA}} \\ &= 26.67 \Omega \end{aligned}$$

- (c) At $V_D = -12 \text{ V}$, $I_D = 1 \mu\text{A}$

$$\begin{aligned} R_D &= \frac{12 \text{ V}}{1 \mu\text{A}} \\ &= 12 \text{ M}\Omega \end{aligned}$$

Observe that the reverse resistance is very high and forward dc resistance decreases with increase in current level.

Example 1.3

Following data are available for a silicon diode.

V_D (Volts)	0.65	0.71	0.76	0.77	0.78	0.79	0.8
I_D (mA)	0	2	4	15	20	25	30

- Determine the ac resistance at $I_D = 2$ mA.
- Determine the ac resistance at $I_D = 25$ mA.
- Calculate the dc resistances at $I_D = 2$ mA and $I_D = 25$ mA and compare with ac resistance levels.

Solution

AC or dynamic resistance is

$$r_d = \frac{\Delta V_d}{\Delta I_d}$$

To find r_d , we have to take symmetrical swing about the specified operating point.

(a) $I_D = 2$ mA

It is convenient to take equal swing of 2 mA about $I_D = 2$ mA since V_D values are provided at $I_D = 0$ mA and 4 mA.

at $I_D = 0$ mA $V_D = 0.65$ V and

at $I_D = 4$ mA $V_D = 0.76$ V

$\therefore$

$$\Delta V_d = 0.76 \text{ V} - 0.65 \text{ V} = 0.11 \text{ V}$$

$$\Delta I_d = 4 \text{ mA} - 0 \text{ mA} = 4 \text{ mA}$$

$$r_d = \frac{0.11 \text{ V}}{4 \text{ mA}} = 27.5 \Omega$$

(b) $I_D = 25$ mA

In this case it is convenient to take equal swing of 5 mA about $I_D = 25$ mA

$V_D = 0.78$ V at $I_D = 20$ mA

$V_D = 0.8$ V at $I_D = 30$ mA

$$\Delta V_d = 0.8 \text{ V} - 0.78 \text{ V} = 0.02 \text{ V}$$

$$\Delta I_d = 30 \text{ mA} - 20 \text{ mA} = 10 \text{ mA}$$

$$r_d = \frac{0.02 \text{ V}}{10 \text{ mA}} = 2 \Omega$$

(c) DC Resistance

$$\text{at } I_D = 2 \text{ mA} \quad R_D = \frac{0.71 \text{ V}}{2 \text{ mA}} = 355 \Omega$$

$$\text{at } I_D = 25 \text{ mA} \quad R_D = \frac{0.79 \text{ V}}{25 \text{ mA}} = 31.6 \Omega$$

The results are compared in the following table.

<i>Diode current</i>	<i>Diode resistance</i>	
	r_d	R_D
2 mA	27.5 Ω	355 Ω
25 mA	2 Ω	31.6 Ω

Note that $r_d \ll R_D$.

Example 1.4

Using the data given in example 1.3, find the average ac resistance when I_D swings from 2 mA to 30 mA.

Solution

$$\begin{aligned} V_D &= 0.71 \text{ V} & \text{at } I_D &= 2 \text{ mA} \\ V_D &= 0.8 \text{ V} & \text{at } I_D &= 30 \text{ mA} \end{aligned}$$

$$\Delta V_d = 0.8 \text{ V} - 0.71 \text{ V} = 0.09 \text{ V}$$

$$\Delta I_d = 30 \text{ mA} - 2 \text{ mA} = 28 \text{ mA}$$

Average ac resistance is

$$\begin{aligned} r_{av} &= \frac{\Delta V_d}{\Delta I_d} \\ &= \frac{0.09 \text{ V}}{28 \text{ mA}} \\ &= 3.2 \Omega \end{aligned}$$

◆ 1.5 ANALYTICAL EXPRESSION FOR DYNAMIC RESISTANCE OF DIODE

In Section 1.4.2 we have illustrated the graphical determination of dynamic resistance of diode. Graphically the dynamic resistance is equal to the reciprocal of the slope of the curve at the Q point. We can also obtain the analytical expression for dynamic resistance using the basic definition in differential calculus.

$$\text{i.e.,} \quad r_d = \left. \frac{dV_D}{dI_D} \right|_{Q\text{point}} \quad (1.9)$$

Let us start with the diode current equation

$$\begin{aligned} I_D &= I_S \left[e^{\frac{V_D}{nV_T}} - 1 \right] \\ I_D &\approx I_S e^{\frac{V_D}{nV_T}} \end{aligned} \quad (1.10)$$

Differentiating with respect to V_D , we obtain

$$\frac{dI_D}{dV_D} = \left[I_S e^{\frac{V_D}{nV_T}} \right] \cdot \frac{1}{nV_T}$$

$$\frac{dI_D}{dV_D} = I_D \left[\frac{1}{nV_T} \right]$$

But
$$\frac{dV_D}{dI_D} = \frac{1}{\left[\frac{dI_D}{dV_D} \right]}$$

$$\therefore r_d = \frac{1}{I_D \left[\frac{1}{nV_T} \right]}$$

or
$$r_d = \frac{nV_T}{I_D} \quad (1.11)$$

Taking $n = 1$ and $V_T = 26 \text{ mV}$ (at room temperature) we have

$$r_d = \frac{26 \text{ mV}}{I_D} \quad (1.12)$$

Note that, the dynamic resistance can be obtained by simply dividing 26 mV by the diode current I_D at Q point.

The following important remarks can be made with reference to Equation (1.12).

- Equation (1.12) gives accurate results only for values of I_D in the vertical-rise section of the curve.
- For lower values of I_D , we have to take $n = 2$ for Si and as a result the value of r_d obtained in Equation (1.12) must be multiplied by 2.
- For small values of I_D below the knee of the curve, Equation (1.12) becomes inappropriate.

1.5.1 Bulk Resistance r_B

The dynamic resistance r_d is the ac resistance of the pn junction. We have not yet considered the following two resistances.

- The resistance of the semiconductor material itself, called the body resistance.
- The resistance introduced by the connection between the semiconductor material and external metallic conductor, called the contact resistance.

The sum of body resistance and contact resistance is called the bulk resistance denoted by r_B .

1.5.2 Dynamic Resistance taking r_B into Account

The dynamic resistance without taking r_B into account is given in Equation (1.12) as

$$r_d = \frac{26 \text{ mV}}{I_D}$$

Now including the effect of r_B , the dynamic resistance is given by

$$\begin{aligned} r'_d &= r_d + r_B \\ r'_d &= \frac{26 \text{ mV}}{I_D} + r_B \end{aligned} \quad (1.13)$$

r_B can range from typically 0.1Ω for high power devices to 2Ω for some low power, general purpose diodes.

At low levels of current, the value of r_B is smaller compared to that of r_d and therefore the effect of r_B can be ignored.

At high levels of current, the value of r_B may approach that of r_d . But the external resistors in the diode circuits will be much larger than r_B and hence effect of r_B is swamped out by these external resistors.

Therefore unless otherwise specified, we ignore r_B in the foregoing analysis.

$$\text{Hence} \quad r'_d \approx r_d = \frac{26 \text{ mV}}{I_D} \quad (1.14)$$

Example 1.5

For the diode considered in example 1.3, calculate the dynamic resistance at $I_D = 2 \text{ mA}$ and 25 mA using Equation (1.12). Compare the results with those obtained in example 1.3 and comment.

Solution

$I_D = 2 \text{ mA}$ [Low current level]

From example 1.3

$$r_d = 27.5 \Omega$$

Using Equation (1.12)

$$r_d = \frac{26 \text{ mV}}{I_D} = 13 \Omega$$

For low current levels, $n = 2$

$\therefore$ The value of r_d calculated using Equation (1.12) has to be multiplied by 2.

$$r_d = 2 [13 \Omega] = 26 \Omega$$

The difference, $27.5 \Omega - 26 \Omega = 1.5 \Omega$, could be treated as contribution due to r_B .

$I_D = 25 \text{ mA}$ [High current level]

From example 1.3

$$r_d = 2 \Omega$$

Using Equation (1.12)

$$r_d = \frac{26 \text{ mV}}{25 \text{ mA}} = 1.04 \Omega$$

For high current levels, $n = 1$

The difference, $2\ \Omega - 1.04\ \Omega = 0.96\ \Omega$, could be treated as contribution due to r_B .

◆ 1.6 EQUIVALENT CIRCUITS OF DIODE

The equivalent circuit of a diode is a circuit that closely approximates the diode behaviour under forward and reverse biased conditions. The diode equivalent circuit is also called the diode model. The analysis of diode circuits can be easily performed using Kirchhoff's voltage law (KVL) and Kirchhoff's current law (KCL), after replacing the diodes with their equivalent circuits.

1.6.1 Piecewise-linear Equivalent Circuit

Piecewise-linear equivalent circuit of a diode can be obtained by approximating the characteristics of the diode by straight line segments as shown in Fig. 1.7.

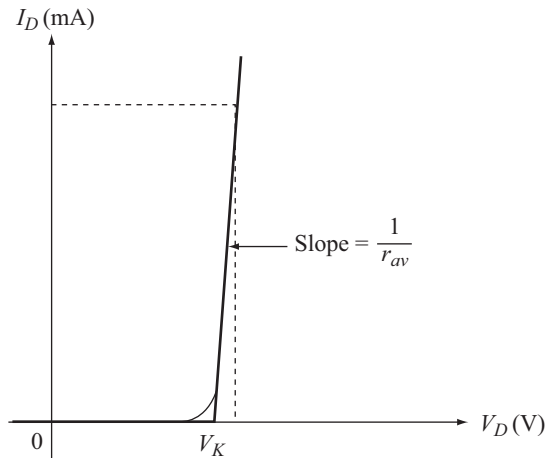


Fig. 1.7 Piece-wise linear characteristics of a diode

We observe the following facts from the piece-wise linear characteristics of Fig. 1.7.

- For $V_D < V_K$, the diode current is zero.
- For $V_D \geq V_K$, the diode current increases linearly with diode voltage. The diode current is related linearly with the diode voltage by the average ac resistance r_{av} .
- The diode acts as an open circuit under reverse bias since, the reverse current through the diode is zero.

Figure 1.8 shows the piecewise linear equivalent circuit of a diode.

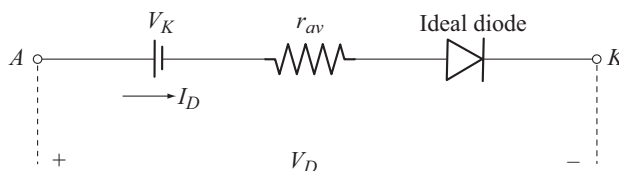


Fig. 1.8 Piecewise linear equivalent circuit of a diode

The polarity of V_K is such that, it opposes the flow of I_D from anode to cathode as long as $V_D < V_K$. Once V_D exceeds V_K , r_{av} limits the diode current.

When conducting, ideal diode acts as a short circuit (0Ω) and behaves as an open circuit ($\infty \Omega$) when it is reverse biased.

1.6.2 Simplified Equivalent Circuit

In most of the diode circuits external resistors will come in series with diodes during conduction. The resistance levels of these elements will be in the order of few hundred ohms to few tens of kilo-ohms. Under this condition r_{av} can be ignored owing to its sufficiently small value. If we apply the approximation, $r_{av} = 0 \Omega$ in the piecewise linear equivalent circuit of Fig. 1.8, we obtain the simplified equivalent circuit of diode shown in Fig. 1.9(b). Since $r_{av} = 0$, slope becomes infinite in the piecewise linear characteristics as shown in Fig. 1.9(a).

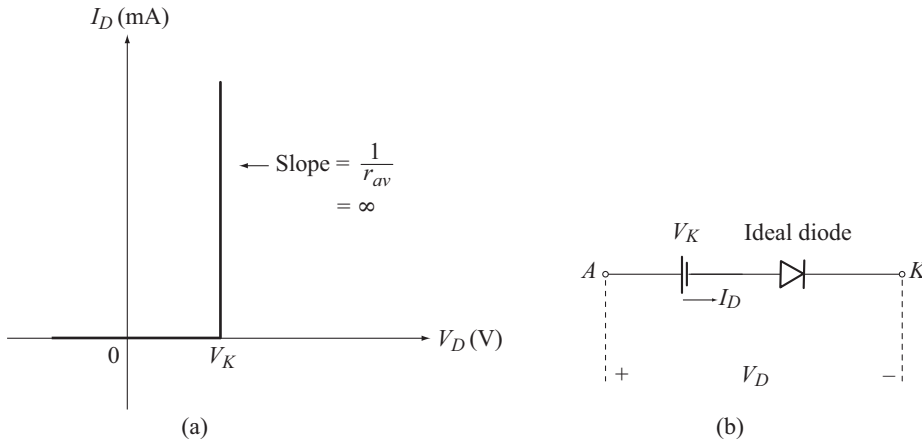


Fig. 1.9 (a) Piece wise linear characteristics with $r_{av} = 0 \Omega$
(b) Simplified equivalent circuit of a diode

1.6.3 Ideal Equivalent Circuit

For Si diode, the level of V_K is 0.7 V. If the external voltage applied to the diode circuit is much greater than V_K , which is true in most of the cases, the effect of V_K can be ignored. If we apply the approximation $V_K = 0$ V to the piecewise linear characteristics and the simplified equivalent circuit of Fig. 1.9 we obtain the piecewise characteristics and the equivalent circuit of ideal diode shown in Fig. 1.10.

Ideal diode is characterised by

- Zero knee voltage ($V_K = 0$ V)
- Zero average ac resistance ($r_{av} = 0 \Omega \Rightarrow$ short circuit between anode and cathode)
- Infinite reverse resistance ($R_r = \infty \Omega \Rightarrow$ open circuit between anode and cathode)

Ideal diode can be regarded as an electronic switch in the sense that, it acts as a short circuit for $V_D \geq 0$ V and open circuit for $V_D < 0$ V.

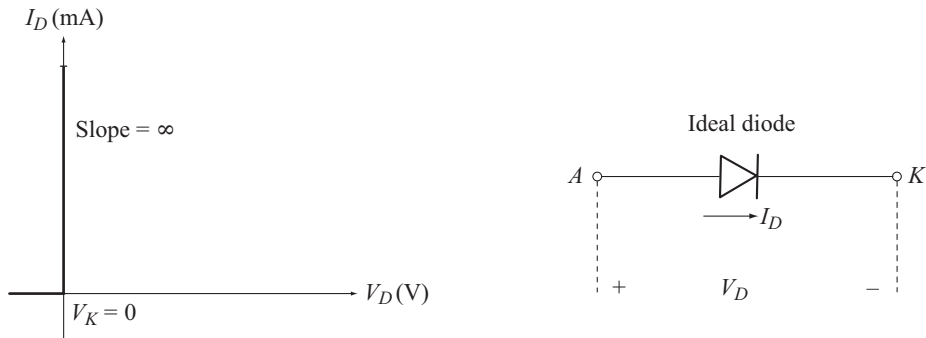
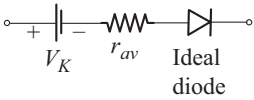
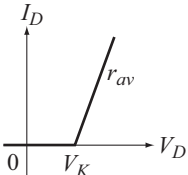
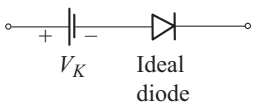
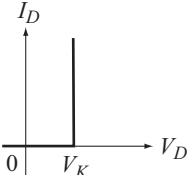
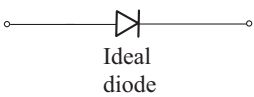
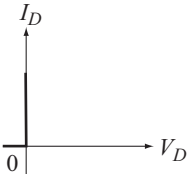


Fig. 1.10 Characteristics and equivalent circuit of ideal diode

1.6.4 Summary of Diode Equivalent Circuits

The summary of diode equivalent circuits is given in Table 1.4.

Table 1.4 Diode equivalent circuits (models)

Type	Conditions	Model	Characteristics
Piecewise-linear model	r_{av} and V_K are considered		
Simplified model	$R_{network} \gg r_{av}$		
Ideal diode	$R_{network} \gg r_{av}$ $E_{network} \gg V_K$		

◆ 1.7 TRANSITION AND DIFFUSION CAPACITANCE

The basic equation for the capacitance of a capacitor is given by

$$C = \frac{Q}{V} \tag{1.15}$$

where Q = Charge on capacitor plate
 V = Applied voltage

In a diode the number of charge carriers crossing the Junction directly depends on the applied voltage, giving rise to a capacitive effect. Two capacitances exist in the diode one in the forward biased region and the other in the reverse biased region. They are

1. Transition or depletion capacitance (C_T)
2. Diffusion or storage capacitance (C_D)

Transition or Depletion Capacitance [C_T]

The basic equation for the capacitance of a parallel plate capacitor is given by

$$C = \frac{A \epsilon}{d} \quad (1.16)$$

A = Plate area

d = Distance between plates

ϵ = Permittivity of the dielectric (insulator) between the plates

In the reverse bias region there exists a depletion region which is free of carriers, that behaves essentially like an insulator between the p and n layers of opposite charge. This gives rise to a capacitance which is called the depletion capacitance or transition capacitance, C_T . With increase in reverse voltage, the depletion width d increases and hence the depletion capacitance decreases as shown in Fig. 1.11(a).

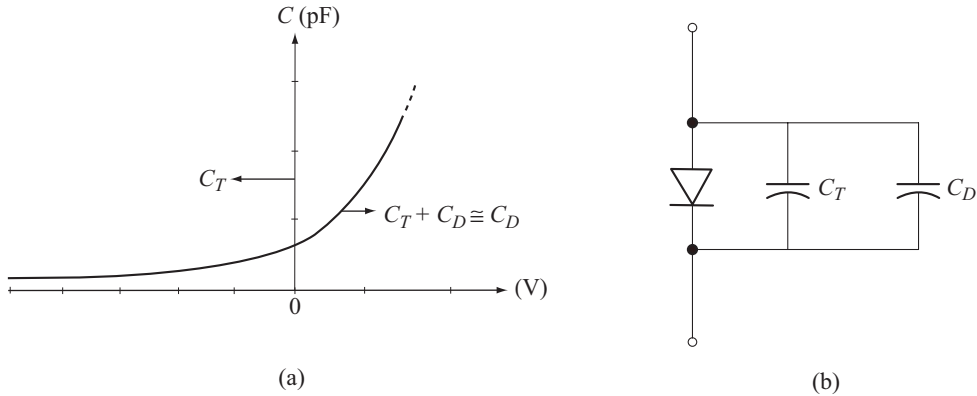


Fig. 1.11 (a) Transition and diffusion capacitance versus applied bias for a Si diode
(b) Diode with its capacitances

Diffusion Capacitance or Storage Capacitance [C_D]

Under forward bias, the holes move from p side to n side and electrons from n side to p side. The electrons in p side and the holes on n side are called minority carriers. The number of excess minority carriers change with the applied bias and constitutes a voltage dependent charge storage or capacitance. Because this extra charge is caused by the diffusion of majority carriers across the junction, the capacitance is called the diffusion capacitance denoted by C_D . With increase in forward voltage, the diffusion of majority carriers will also increase giving rise to an increase in diffusion capacitance as shown in Fig. 1.11(a).

It is important to note that, depletion capacitance is dominant under reverse bias. Since the depletion width is very small under forward bias, diffusion capacitance is predominant. The effect of C_T and C_D are considered by placing them in parallel across the diode as shown in Fig. 1.11(b).

Special diodes optimised for use as voltage-dependent capacitors, operating under reverse bias are called varactor diodes. Varactor diodes find application in communication systems.

◆ 1.8 CHARGE STORAGE AND REVERSE RECOVERY TIME

Consider a rectifier diode which is forward biased and carrying a constant forward current I_F as shown in Fig. 1.12.

Due to the application of forward bias, holes move from p region to n region and electrons move from n region to p region (it may be noted that holes in n region and electrons in p region are minority carriers). Therefore when the diode is conducting, p region has plenty (excess) of electrons and n region has plenty (excess) of holes which are taking part in conduction. The excess electrons in p region and excess holes in n region are called stored charges and this phenomenon is called charge storage.

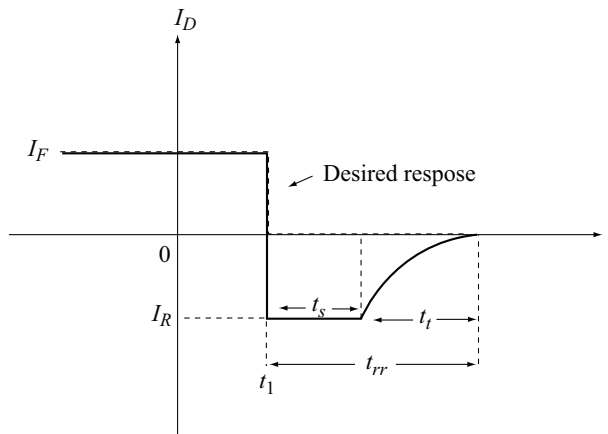


Fig. 1.12 Reverse recovery in rectifier diode

If the conducting diode is suddenly reverse biased at $t = t_1$, the diode current does not fall immediately from I_F to zero but it becomes zero after a time, t_{rr} , called reverse recovery time as explained below.

1. A large number of stored charge, (minority carriers) are present in each region. A certain amount of time, t_s , called the storage time is required for the minority carriers to return to their majority carrier state. Since excess holes in n region are moving towards p region and excess electrons in p region are moving towards n region, the current direction reverses in the diode. As a result the diode conducts a reverse current I_R during the interval t_s .
2. Once the excess or stored charges returned to their original regions, the reverse current gradually decreases from I_R to zero in the interval t_t , which is called the transition interval.

The reverse recovery time t_{rr} is the sum of storage time t_s and the transition time t_t .

$$\text{i.e.,} \quad t_{rr} = t_s + t_t \quad (1.17)$$

As a guide, t_{rr} is usually defined as the time taken by the reverse current to drop to 10 percent of the forward current.

For example IN4148 has a t_{rr} of 4 ns. If this diode has a forward current of 20 mA, and it is suddenly reverse biased, it takes approximately 4 ns for the reverse current to decrease to 2 mA.

◆ 1.9 DIODE SPECIFICATION SHEETS

Specification sheets or data sheets for semiconductor devices are provided by the manufacturer, which give certain data or characteristics that are very helpful to the designer while selecting the devices for a specific applications. The commonly provided data for a diode on specification sheets are:

1. The forward voltage V_F (at a specified current and temperature)
2. The maximum forward current I_F (at a specified temperature)
3. The reverse saturation current I_R (at a specified voltage and temperature)
4. The reverse-voltage rating PIV or PRV (at a specified temperature)
5. The maximum power dissipation level (at a particular temperature)

$$P_{D \max} = V_D I_D \\ \approx (0.7 \text{ V}) I_D \quad [\text{For Si diode}]$$

6. Capacitance levels
7. Reverse recovery time (t_{rr})
8. Operating temperature range.

For the silicon diode BAY73 we find the following data on specification sheets.

1. Maximum forward current I_F : 500 mA
2. Reverse saturation current I_R : 500 nA
3. Peak inverse voltage PIV: 100 V
4. Break down voltage: 125 V
5. Maximum power dissipation $P_{D \max}$: 500 mW
6. Capacitance: 8 pF
7. Reverse recovery time t_{rr} : 3 μ S
8. Operating temperature range: 175 °C (maximum)

For instance, if in a given application the expected maximum reverse voltage is 50 V, BAY 73 diode can be safely used since its PIV rating is 100 V. If the expected maximum reverse voltage is more than 100 V, then the diode with higher PIV rating must be selected. Similarly the other ratings must be considered while selecting a diode for a specified application.

◆ 1.10 CLASSIFICATION OF JUNCTION DIODES

Based on currents, diodes may be classified as

- (i) Low-current diodes
- (ii) Medium-current diodes
- (iii) High-current diodes

Typically low-current diodes have a body 3 mm long with a coloured band close to the cathode for identification as shown in Fig. 1.13. These diodes can withstand forward currents of the order of 100 mA and reverse voltage of about 75 V. The reverse current at room temperature is typically less than 1 μ A.

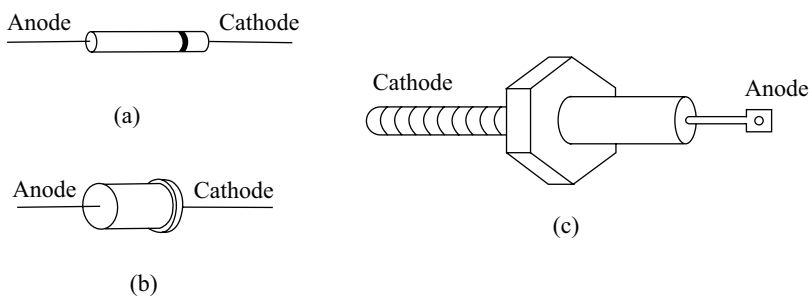


Fig. 1.13 (a) Low current diode
(b) Medium current diode
(c) High current diode

Medium current diodes are packaged in a metal can-like casing of typical diameter 8.9 mm and length 7.6 mm as shown in Fig. 1.13(b). They can carry a forward current in the range of about 400 – 500 mA and withstand reverse voltage of about 200 – 300 V.

The appearance of a high current diode in a solid metal package is shown in Fig. 1.13(c). The diode is screwed into a metal heat sink for quick dissipation of the heat generated when high current is conducted through the device. Typical body diameter is 7.8 mm and the total length is 31.2 mm. These diodes can carry several amperes of current and can withstand reverse voltage of several hundreds of volts.

◆ 1.11 LOAD-LINE ANALYSIS OF DIODE CIRCUIT

The load line analysis is the graphical method of analysing diode circuits. In this method the current and voltage levels in the diode circuit to the applied load are obtained by drawing the load line on the actual diode characteristics. The procedure for constructing the load-line is explained below.

Figure 1.14(a) shows a series diode circuit and characteristics of diode is shown in Fig. 1.14(b). Applying KVL to the circuit of Fig. 1.14(a) we have

$$V - V_D - I_D R = 0$$

or

$$V = V_D + I_D R \quad (1.18)$$

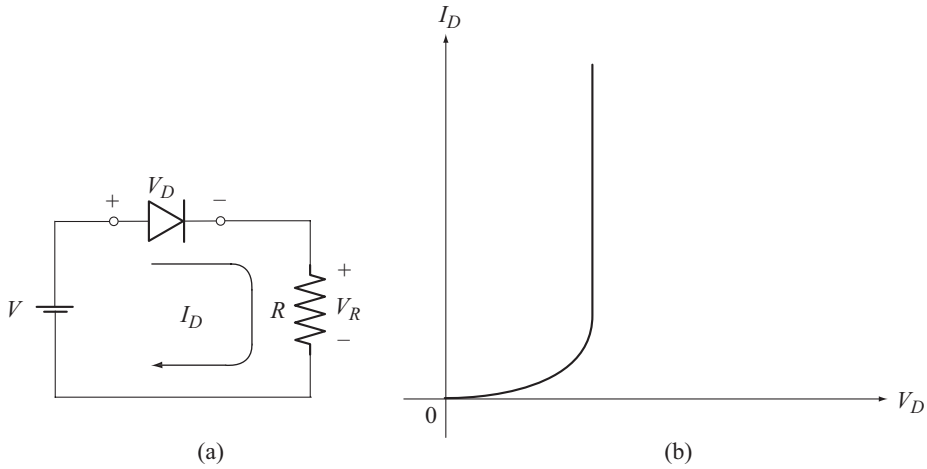


Fig. 1.14 (a) Series diode configuration
(b) Diode characteristics

For any point say A on I_D axis, $V_D = 0$.
Using $V_D = 0$ in Equation (1.18) we have

$$I_D = \frac{V}{R} \quad (1.19)$$

Now the coordinates of point A are

$$A(V_D, I_D) = A\left(0, \frac{V}{R}\right)$$

Similarly for any point B on V_D axis, $I_D = 0$.
Using $I_D = 0$ in Equation (1.18), we obtain

$$V_D = V \quad (1.20)$$

Now the coordinates of point B are

$$B(V_D, I_D) = B(V, 0)$$

Let us draw the load line on the diode characteristics by connecting the points A and B as shown in Fig. 1.15. The intersection of load line with characteristics curve gives the Q point.

The co-ordinates of the Q point, gives the diode voltage V_{DQ} and the diode current I_{DQ} for the applied load R .

The straight line AB is also called the dc load line since it is drawn for the dc conditions. Once I_{DQ} is known we can find the voltage across the load resistance R using

$$V_R = I_{DQ} R \quad (1.21)$$

or

$$V_R = V - V_{DQ} \quad (1.22)$$

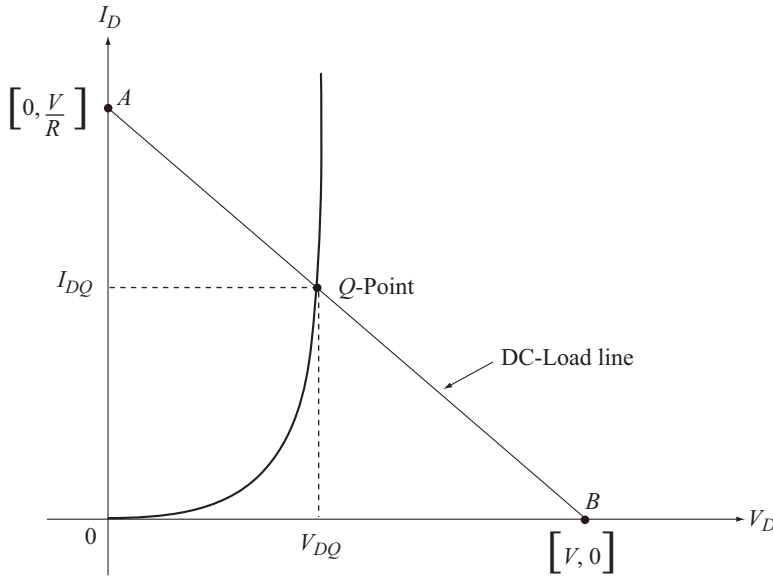


Fig. 1.15 DC load-line drawn on diode characteristics

Equation (1.18) can also be put in the form

$$I_D = \left[-\frac{1}{R} \right] V_D + \frac{V}{R} \quad (1.23)$$

Equation (1.23) represents the dc load line with a slope $-\frac{1}{R}$ and intercept $\frac{V}{R}$ on I_D axis.

Load line analysis described above uses the diode characteristics which is available on specification sheets. The characteristics of the device in actual use may be different from that available on data sheet. Hence the results obtained may be slightly different from the actual results. Also the graphical analysis become tedious for a complex circuit which uses diodes in many branches. The best method to analyse the diode circuits is to use the diode equivalent circuit which is described in the next section.

◆ 1.12 ANALYSIS OF DIODE CIRCUITS USING THE DIODE EQUIVALENT CIRCUIT

In this technique each diode in the circuit is replaced by its appropriate equivalent circuit depending upon whether the diode is forward biased or reverse biased. The diode equivalent circuits were developed in section 1.6.1 using the piece - wise linear characteristics.

In majority of the circuits, the external circuit resistance levels are much greater than the average diode resistance. Hence we can use approximate equivalent circuit which is reproduced in Fig. 1.16(a), neglecting the effect of r_{av} .

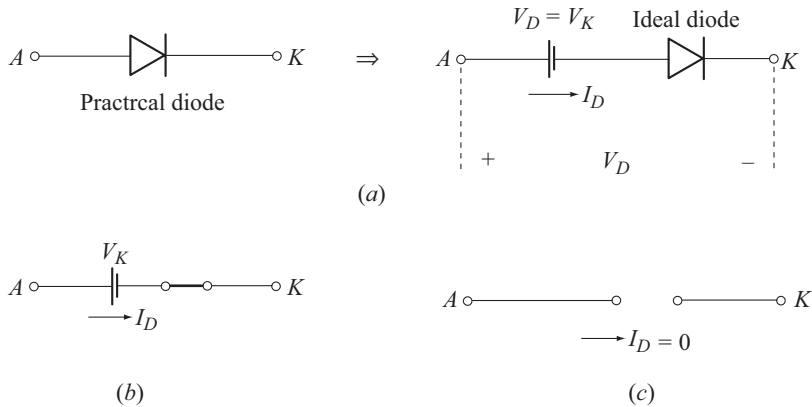


Fig. 1.16 (a) Approximate equivalent circuit of diode
 (b) Equivalent circuit when diode conducts
 (c) Equivalent circuit when diode is reverse biased

Note that ideal diode represents short circuit when it conducts and an open circuit under reverse bias as shown in Fig. 1.16(b) and Fig. 1.16(c) respectively.

If V_K is neglected compared with external applied voltages the equivalent circuit reduces to that of ideal diode which represents a short circuit under forward bias and open circuit under reverse bias as shown in Fig. 1.17.

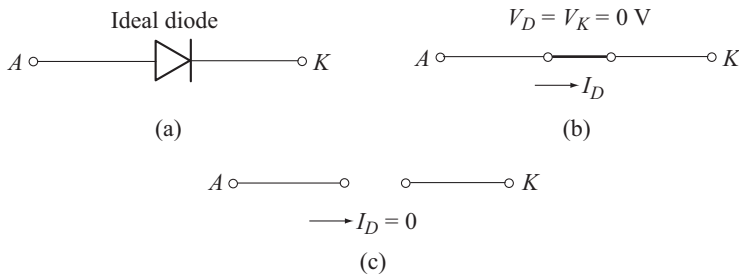


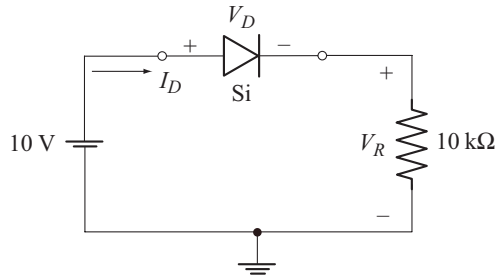
Fig. 1.17 (a) Ideal diode
 (b) Equivalent circuit under forward bias
 (c) Equivalent circuit under reverse bias

The analysis of series, parallel and series-parallel diode circuits are considered in the following examples.

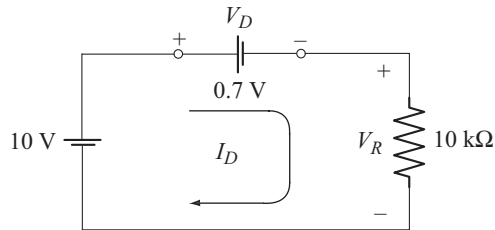
Example 1.6

For the diode circuit shown

- Calculate I_D , V_D and V_R
- Repeat part (a) if the battery polarity is reversed

**Solution**

- (a) The diode conducts since the battery voltage forward biases the diode ($10\text{ V} > V_K$). Let us replace the diode by its approximate equivalent circuit.



$$V_D = V_K = 0.7\text{ V}$$

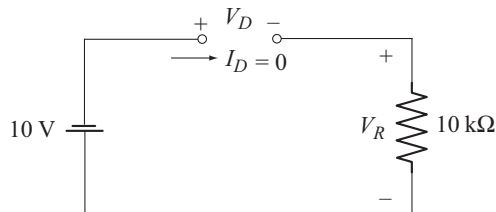
Applying KVL we have

$$10\text{ V} - 0.7\text{ V} - I_D [10\text{ k}\Omega] = 0$$

$$I_D = \frac{9.3\text{ V}}{10\text{ k}\Omega} = 0.93\text{ mA}$$

$$\begin{aligned} V_R &= I_D (10\text{ k}\Omega) \\ &= (0.93\text{ mA}) (10\text{ k}\Omega) \\ &= 9.3\text{ V} \end{aligned}$$

- (b) If the battery polarity is reversed, the diode gets reverse biased and hence it acts as an open circuit as shown below.



$$I_D = 0\text{ mA}$$

$$V_R = I_D (10\text{ k}\Omega) = 0\text{ V}$$

Writing KVL equation we get

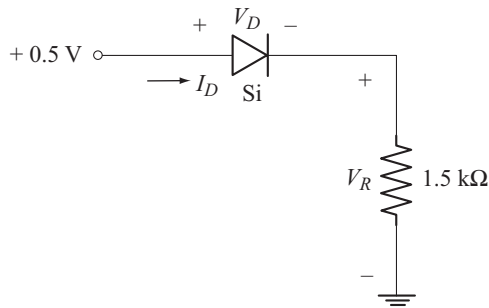
$$-10 \text{ V} - V_D - I_D (10 \text{ k}\Omega) = 0$$

$$\Rightarrow V_D = -10 \text{ V}$$

Note that the diode is reverse biased to the level of -10 V .

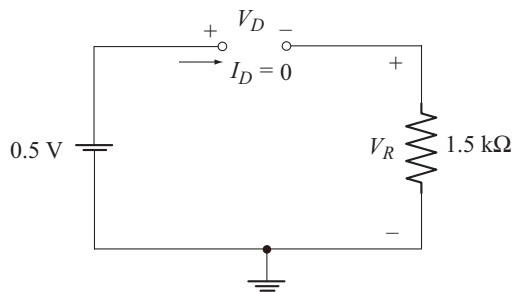
Example 1.7

For the circuit shown find I_D , V_D and V_R .



Solution

The applied forward voltage is only 0.5 V which is less than $V_K = 0.7 \text{ V}$. Hence the diode does not conduct and it acts as an open circuit as shown below.



$$I_D = 0 \text{ mA}$$

$$V_R = I_D (1.5 \text{ k}\Omega) = 0 \text{ V}$$

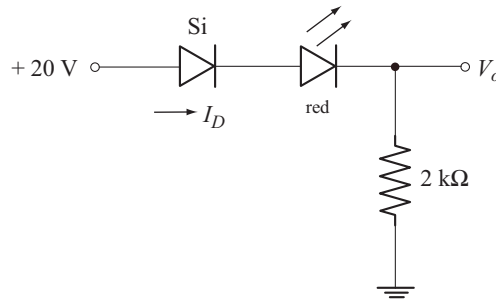
Applying KVL we get

$$0.5 \text{ V} - V_D - I_D (1.5 \text{ k}\Omega) = 0$$

$$\Rightarrow V_D = 0.5 \text{ V}.$$

Example 1.8

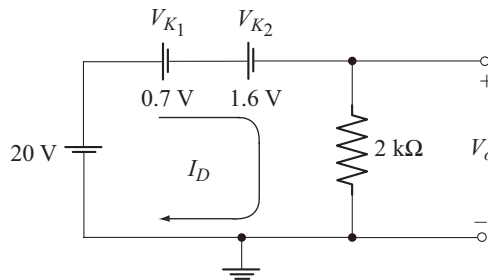
For the circuit shown below determine V_o and I_D . Assume $V_K = 1.6$ V for the red LED.

**Solution**

Both diodes conduct in the same direction. Since both are in series, the total knee voltage is

$$\begin{aligned} V_K &= V_{K_1} + V_{K_2} \\ &= 0.7 \text{ V} + 1.6 \text{ V} \\ &= 2.3 \text{ V.} \end{aligned}$$

Both diodes are turned on since the applied voltage is more than 2.3 V. The equivalent circuit is shown below.

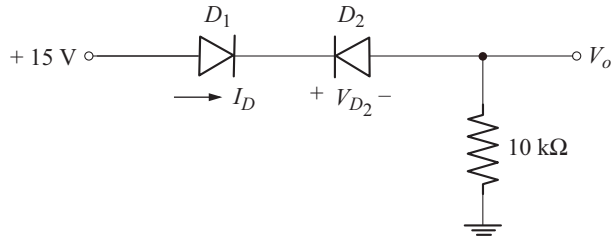


Applying KVL we have

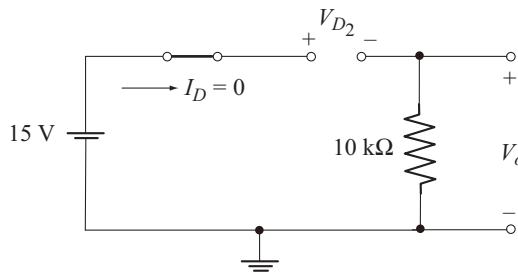
$$\begin{aligned} 20 \text{ V} - 0.7 \text{ V} - 1.6 \text{ V} - I_D (2 \text{ k}\Omega) &= 0 \\ I_D &= \frac{17.7 \text{ V}}{2 \text{ k}\Omega} = 8.85 \text{ mA} \\ V_D &= I_D (2 \text{ k}\Omega) \\ &= (8.85 \text{ mA}) (2 \text{ k}\Omega) \\ &= 17.7 \text{ V} \end{aligned}$$

Example 1.9

For the circuit shown below determine I_D , V_{D_2} and V_o . D_1 and D_2 are both silicon diodes.

**Solution**

The applied voltage forward biases D_1 but reverse biases D_2 . The current in the circuit is zero since D_2 acts as an open circuit. Though D_1 is forward biased, its current is zero. Hence we have to replace D_1 by short circuit instead of V_K . The equivalent circuit is shown below.



$$I_D = 0 \text{ mA}$$

$$V_o = I_D (10 \text{ k}\Omega) = 0 \text{ V}$$

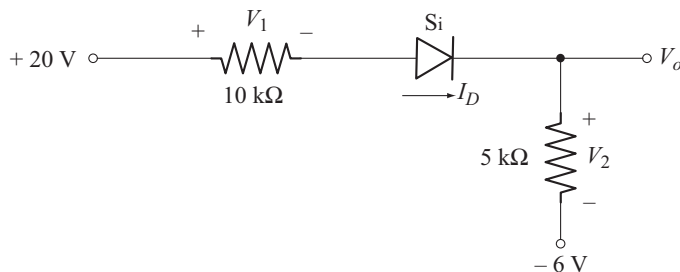
Applying KVL we have

$$15 \text{ V} - V_{D_2} - V_o = 0$$

$$V_{D_2} = 15 \text{ V}$$

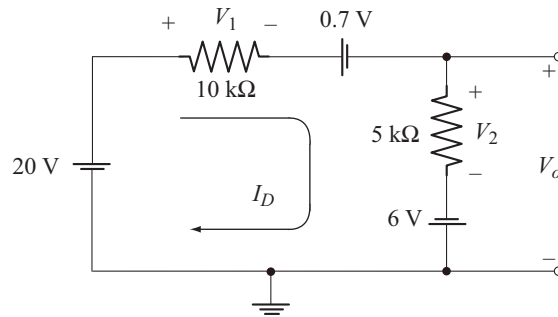
Example 1.10

For the circuit shown below determine I_D , V_1 , V_2 and V_o .



Solution

The diode is in the on state since the anode has positive potential and the cathode negative potential. The equivalent circuit is shown below.

**Fig. A**

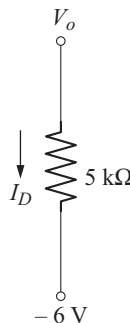
Applying KVL to the circuit of Fig. A, we get

$$20\text{ V} - 10\text{ k}\Omega (I_D) - 0.7\text{ V} - 5\text{ k}\Omega (I_D) + 6\text{ V} = 0$$

$$I_D = \frac{26\text{ V} - 0.7\text{ V}}{10\text{ k}\Omega + 5\text{ k}\Omega} = 1.68\text{ mA}$$

$$V_1 = I_D (10\text{ k}\Omega) = (1.68\text{ mA}) (10\text{ k}\Omega) = 16.8\text{ V}$$

$$V_2 = I_D (5\text{ k}\Omega) = (1.68\text{ mA}) (5\text{ k}\Omega) = 8.4\text{ V}$$

**Fig. B**

From the circuit of Fig. B

$$I_D = \frac{V_o - (-6\text{ V})}{5\text{ k}\Omega}$$

$\Rightarrow$

$$V_o = I_D (5\text{ k}\Omega) - 6\text{ V} = (1.68\text{ mA}) (5\text{ k}\Omega) - 6\text{ V} = 2.4\text{ V}$$

Alternatively, applying KVL to the path consisting of V_o , V_2 and 6 V battery we have

$$V_o - V_2 + 6 \text{ V} = 0$$

$$V_o = V_2 - 6 \text{ V} = 8.4 \text{ V} - 6 \text{ V} = 2.4 \text{ V}$$

Example 1.11

For each of the following circuits determine the current I . Assume silicon diodes.

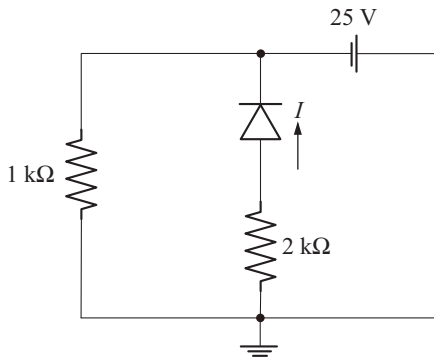


Fig. A

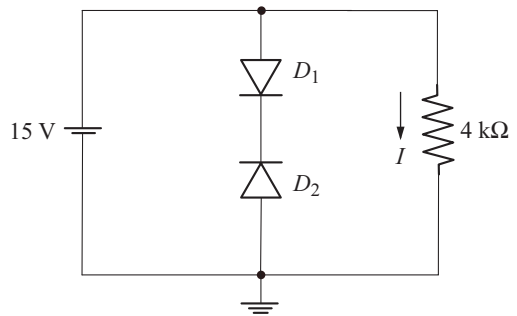
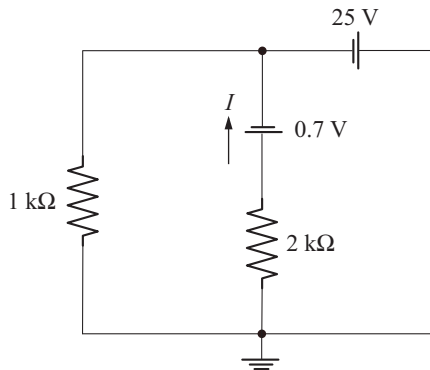


Fig. B

Solution

To find I in the circuit of Fig. A

The diode is in the on state. The equivalent circuit is shown below.



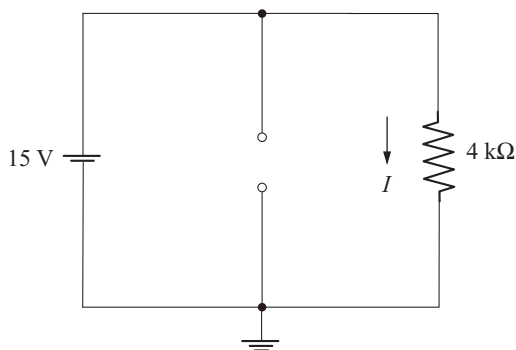
Applying KVL to the second loop we have

$$-I(2 \text{ k}\Omega) - 0.7 \text{ V} + 25 \text{ V} = 0$$

$$I = \frac{25 \text{ V} - 0.7 \text{ V}}{2 \text{ k}\Omega} = 12.15 \text{ mA}$$

To find I in the circuit of Fig. B

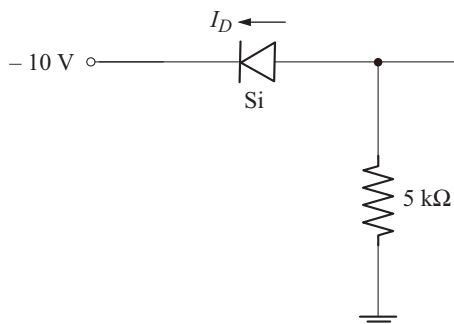
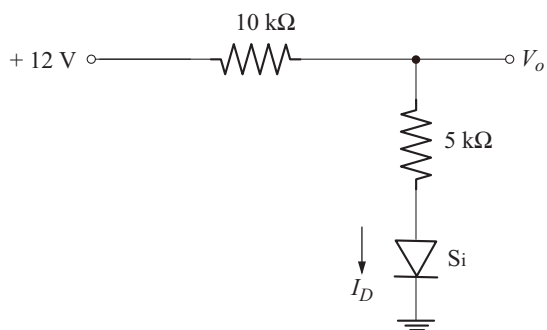
Since D_2 is reverse biased, the diode branch can be treated as an open circuit. The equivalent circuit is shown below.



$$I = \frac{15 \text{ V}}{4 \text{ k}\Omega} = 3.75 \text{ mA}$$

Example 1.12

Determine V_o and I_D for the circuits shown below.

**Fig. A****Fig. B****Solution**

To find I_D and V_o for the circuit of Fig. A

The diode is in the on state. The equivalent circuit is shown below.

